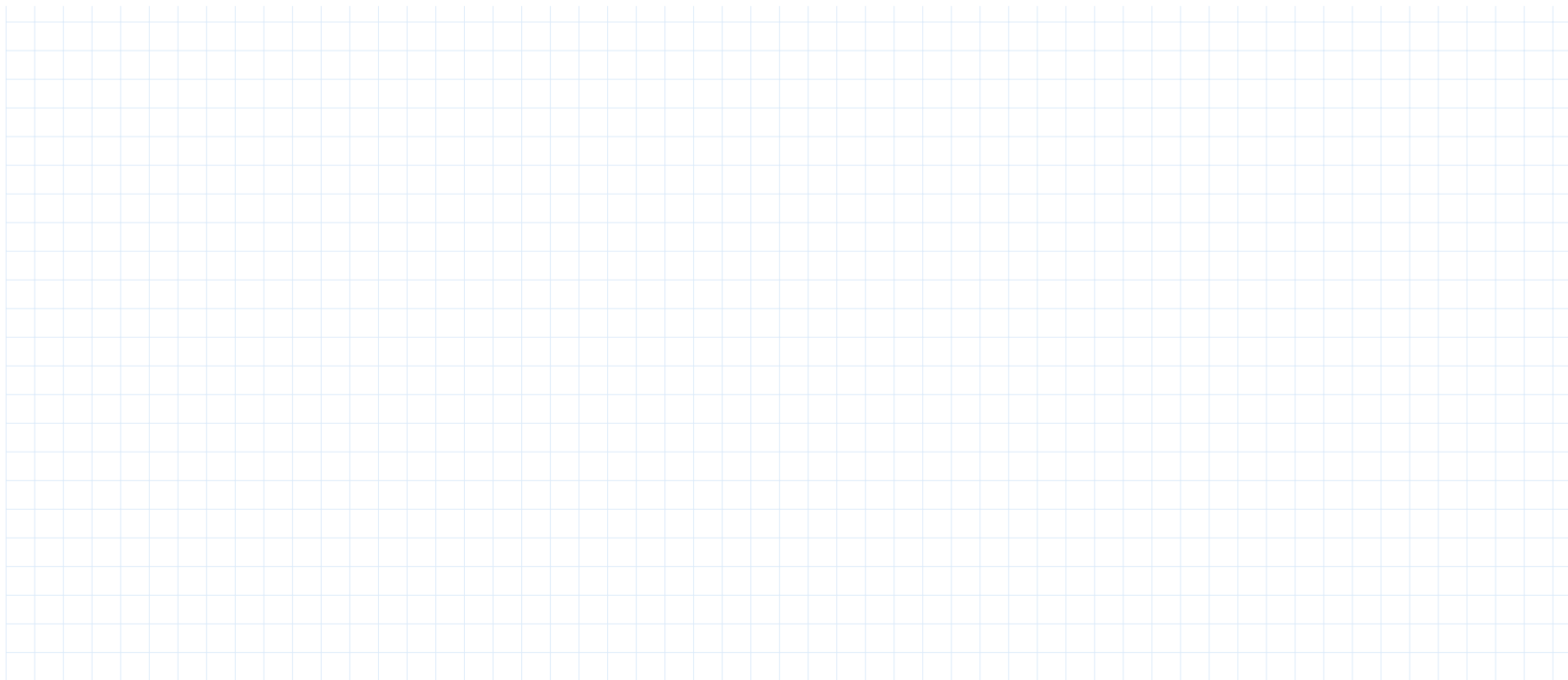


Angles and radians

An angle in **standard position** starts on the positive x -axis. Counterclockwise rotation is positive; clockwise is negative. A half-turn is $180^\circ = \pi$ radians.

Example. Convert 225° to radians and $-\frac{5\pi}{6}$ to degrees. Sketch each angle in standard position.

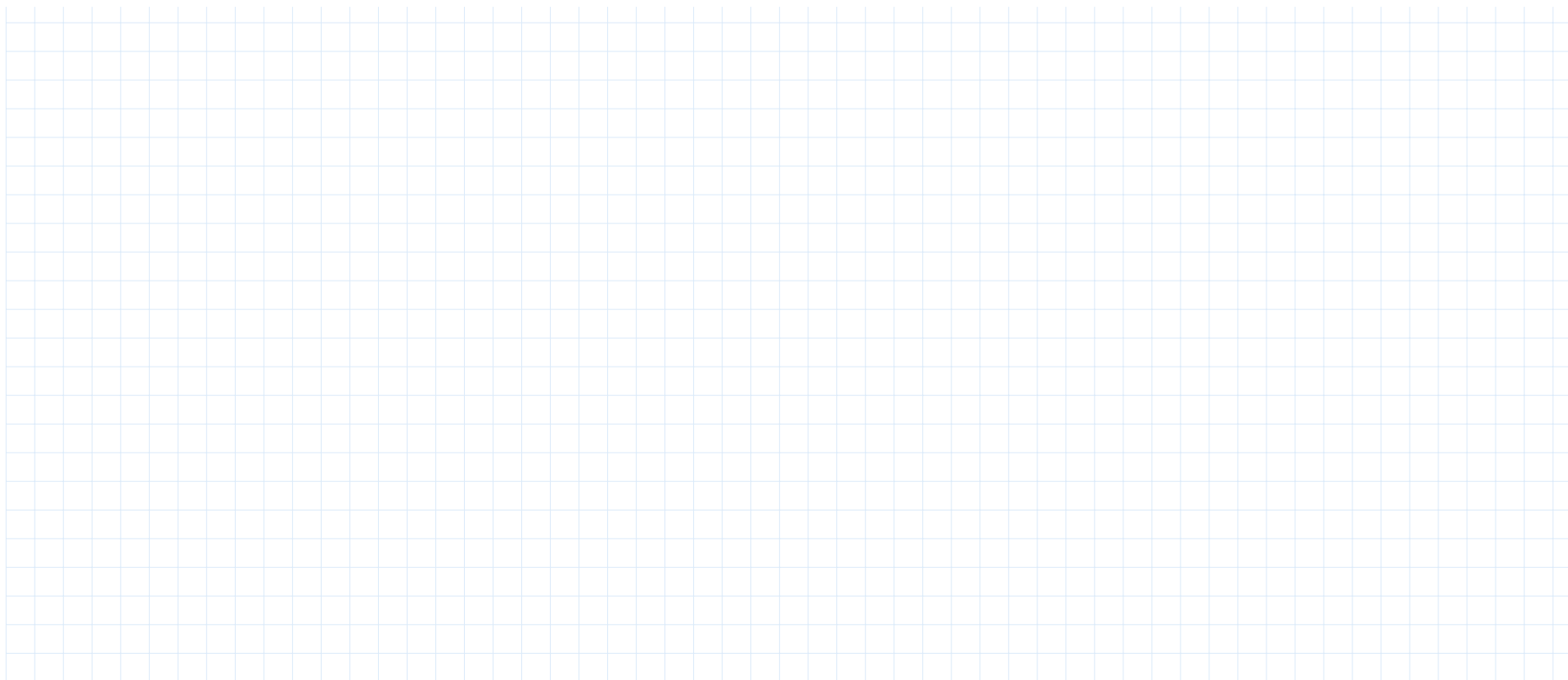


Degrees, minutes, seconds

One degree contains 60 minutes; one minute contains 60 seconds:

$$1^\circ = 60', \quad 1' = 60''.$$

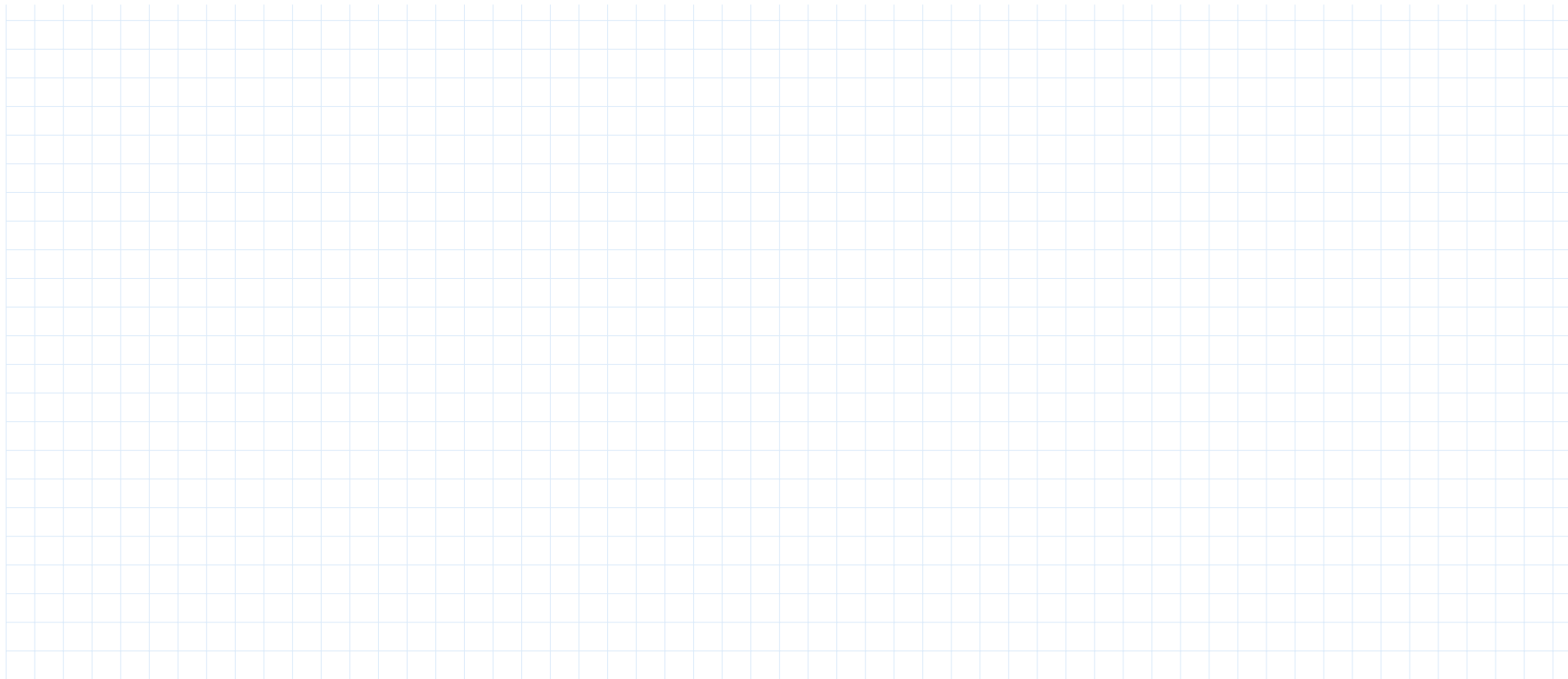
Example. (a) Write $37^\circ 18' 36''$ as a decimal number of degrees. (b) Write 82.425° in degrees, minutes, and seconds.



Arc length

For a circle of radius r , an angle of θ **radians** cuts off an arc of length $s = r\theta$.

Example. A circle has radius 8 cm. Find the exact length of the arc subtended by an angle of 135° . Sketch the arc.

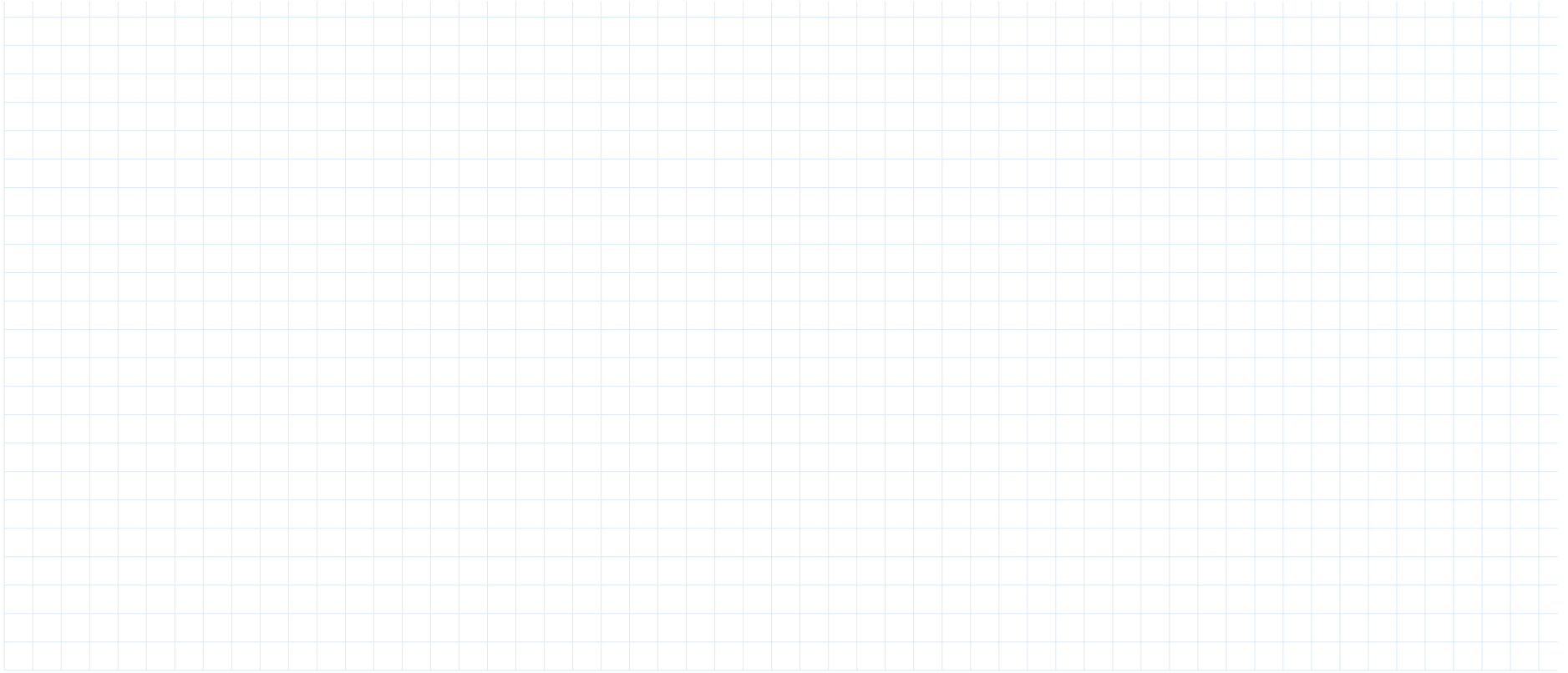


Area of a sector

A sector is the region swept out by a central angle. If θ is in **radians**, then

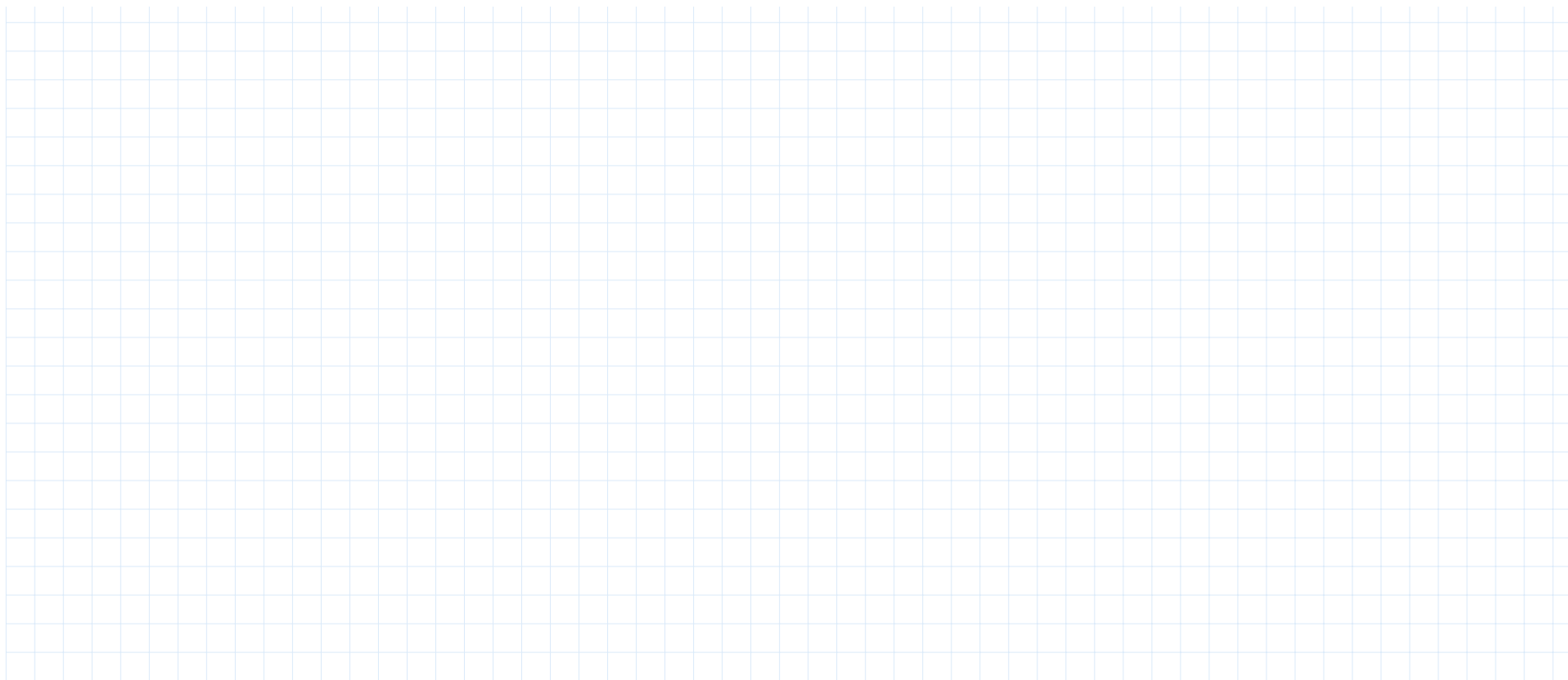
$$A = \frac{1}{2}r^2\theta.$$

Example. A sector has radius 10 m and area 25π m². Find its central angle in radians and in degrees.



Distance along a circle

Example. Two locations are due north and south of each other on Earth. Their latitudes are $15^{\circ}20'$ N and $22^{\circ}50'$ N. Assuming Earth's radius is 3960 miles, estimate their surface distance to the nearest mile.



For motion around a circle,

$$\omega = \frac{\theta}{t} \quad (\text{radians per unit time}), \quad v = r\omega.$$

Example. A wheel of radius 0.4 m rotates at 45 revolutions per minute. Find its angular speed in radians per minute and its rim speed in meters per minute.

